

Yap Hong Ping

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Applied AI Engineer | Production AI Systems · Full-Stack · LLM/RAG · Data Pipelines

Professional Summary

Applied AI Engineer with 10+ years building and shipping production-grade AI systems end-to-end — from ambiguous problem statements through data pipelines, model development, and full-stack deployment. Specializes in agentic AI, RAG orchestration, and NLP with FastAPI backends, delivering MVPs within 2 weeks using AI-native development tools (GitHub Copilot SpecKit) and automated CI/CD. Proven ability to translate business requirements into production applications with measurable investment in operational efficiency.

Core Skills

AI/ML: Agentic Workflows, RAG, LLM Orchestration, NLP, Computer Vision, ML Pipelines

Backend: Python (10+ yrs), FastAPI, Django-familiar, REST APIs, Microservices

LLM/GenAI: LangChain, Prompt Engineering, Context Engineering, Multi-Agent Pipelines, Embeddings

Data Pipelines: ETL/ELT, Data Extraction, Feature Engineering, Elasticsearch, Hadoop

Databases: Elasticsearch (vector DB), Oracle, MSSQL, MySQL, PostgreSQL-familiar

Cloud & Infra: Docker, Kubernetes, OpenShift, AWS (SageMaker), GCP (Vertex AI), Azure

MLOps/DevOps: GitLab CI/CD, ArgoCD (GitOps), Helm, Artifactory, GitHub Actions

AI-Native Dev: GitHub Copilot (SpecKit), spec-driven development, AI-augmented workflows

Evaluation: Model evaluation frameworks, F1/precision/recall, latency benchmarking, A/B metrics

Professional Experience

Infineon Technologies Asia Pacific — Staff Analyst, Data Scientist

Nov 2021 – Present | Singapore

Multi-Agent AI System – Production Application with Full CI/CD

- Translated ambiguous business requirements into production-grade multi-agent AI application, delivering deployed MVP in 2 weeks with spec-driven development and automated CI/CD pipeline
- Designed and implemented full-stack application: FastAPI backend + Streamlit frontend + hierarchical agentic pipeline (Planner → Sub-Agents → Checker) orchestrating 4 enterprise data sources

- Built data processing workflows: automated data extraction, transformation, and integration pipelines provisioning 4 knowledge sources with source-cited responses
- Implemented evaluation framework measuring response quality, latency (~5sec), and user adoption metrics; deployed across dev/staging/prod environments via GitOps
- Designed complete CI/CD: GitLab CI (test/lint/build/deploy) → OpenShift BuildConfig → Artifactory → ArgoCD with Helm charts and environment-specific overlays

Tech: Python, LangChain, FastAPI, Streamlit, GitLab CI/CD, ArgoCD, OpenShift, Docker, Helm, Elasticsearch

Enterprise RAG Platform – Knowledge Retrieval System (Production)

- Designed end-to-end data pipeline: ingestion → document parsing → chunking → embedding → vector indexing across ~1,000 documents (ppt/doc/xlsx/pdf/html)
- Built retrieval-augmented generation system with tool-calling integration, reducing information search time from ~2 hours to ~5 minutes
- Achieved ~100 queries/day adoption; implemented monitoring and evaluation metrics for response quality and latency

Tech: Python, LangChain, Elasticsearch (vector DB), FastAPI, FastMCP, Docker, Kubernetes

NLP Classification Pipeline – Document Intelligence (Production)

- Developed end-to-end ML pipeline: OCR + layout understanding → feature engineering → fine-tuned transformer classification across 11 topics (F1=0.8)
- Built data extraction and transformation workflow processing 200+ labelled contracts with human-in-the-loop validation for continuous improvement

Tech: Python, HuggingFace, PyTorch, EasyOCR, FastAPI

Computer Vision & OCR – Receipt Auditing System (POC)

- Built image classification + segmentation + OCR pipeline on ~1,000 labelled images with rule-based validation and event-trigger integration to SAP BTP

PSA Corporation — Principal Engineer, Smart Systems Solutions

Jul 2018 – Nov 2021 | Singapore

- Delivered predictive maintenance system for 20+ machines (0.5s sampling): data pipeline → anomaly detection → classification → dashboard; saved 8 man-hours/week
- Built object detection models for equipment monitoring; partnered with vendors for production deployment
- Designed data pipelines on Cloudera Hadoop Platform for industrial sensor data processing

Tech: Python, Scikit-learn, PyTorch, DataRobot, Cloudera Hadoop

PSA Corporation — Senior Mechanical Engineer

Jul 2013 – Jul 2018 | Singapore

- Developed automated component life tracking systems; translated mechanical failure patterns into rule-based monitoring criteria

Portfolio Projects

Project	Description	Platform
receipt-ai	OCR + model serving endpoint	AWS SageMaker
rag-chatbot	RAG + agentic tool-use + open-source LLMs	Azure
contract-clause-extractor	Legal NLP + Streamlit demo	GitHub

Education

Master of Technology (Knowledge Engineering) — NUS-ISS | 2018–2020

Bachelor of Engineering (Mechatronic Engineering) — NTU | 2009–2013 | Second Class Honours (Upper)